

Free energy with divisible perturbations

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Abstract

Unit 66's asymptotic for $\xi = a + u^k \tilde{J}$, $\eta = \eta_0 + u^k \tilde{\eta}$ combined with the logarithm lemma of unit 61 gives $-\log Z(\xi, \eta) = \frac{p}{2} \log n - m \log \log n + F_0 + o(1)$ for $\eta_0 > 0$, with $F_0 = -\log(\eta_0 C)$: the perturbations move only the constant term of the chart free energy. Zero `sorry/axiom`.

1 The things formalised

```
theorem boxIntegralXiEta_freeEnergy ... (h : 0 < ) :
  (p : ) (m : ) (F : ), ( i, p finExp k h i)
  m + 1 = (Finset.univ.filter fun i => finExp k h i = p).card
  Tendsto (fun n => -Real.log (boxIntegralXiEta a b (Real.sqrt n) k h
    (fun u => ( i, u i ^ k i) * J u) (fun u => + ( i, u i ^ k i) * u))
    - (p / 2 * Real.log n - m * Real.log (Real.log n))) atTop ( F )
```

2 Proof strategy

`tendsto_log_of_isEquivalent_powLog` applied to unit 66's equivalence with constant $\eta_0 C > 0$, negated and rearranged.

3 Roadblocks and resolutions

- None: first-pass compile.

4 Outcome

The chart free energy has the Watanabe form $\lambda \log n - (m - 1) \log \log n + O(1)$ for the whole class of divisible perturbations of ξ and η , with λ and m read off from (k, h) .